



Hamilton School for
Classical and Civic Education
UNIVERSITY *of* FLORIDA

The State of Nature

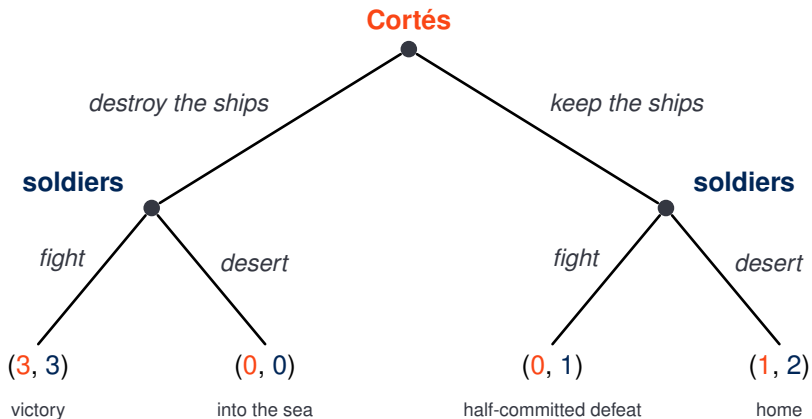
An Introduction to Game Theory
ISS 2440 — Intro to PPEL

Louis Rouanet

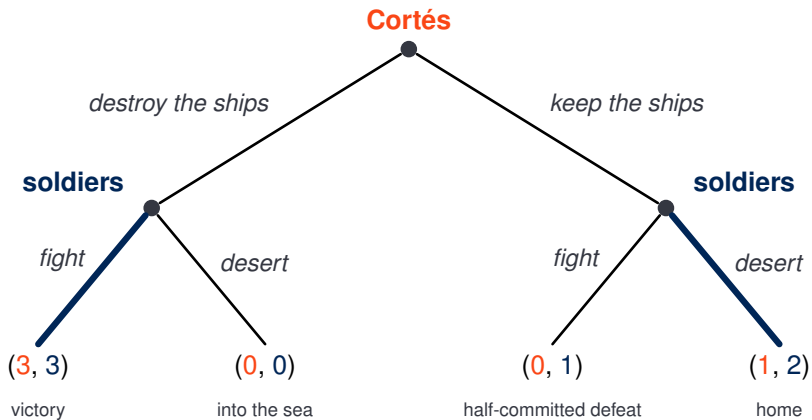
Fall 2026



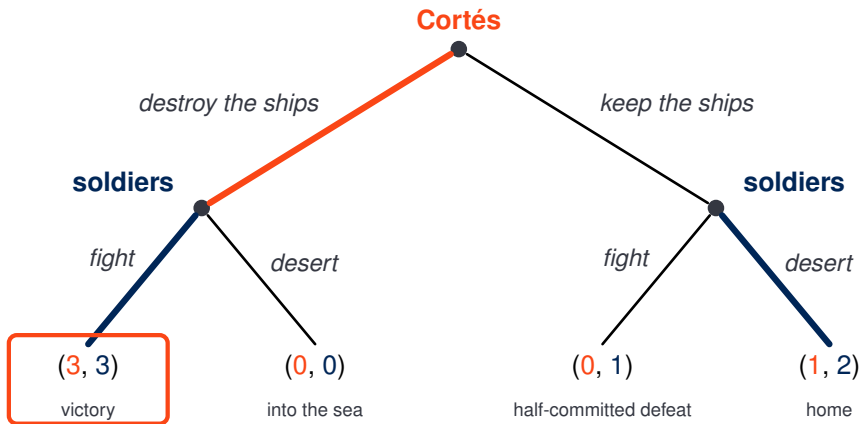
UF | The game Cortés played



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- ▶ Solve the game from the end: in each stage-2 world, find the soldiers' best reply first.
- ▶ Then move up: Cortés, **anticipating** those replies, compares 3 against 1 and destroys the ships.
- ▶ Reasoning from the last move back to the first is called **backward induction**. It is how every game tree in this course gets solved.

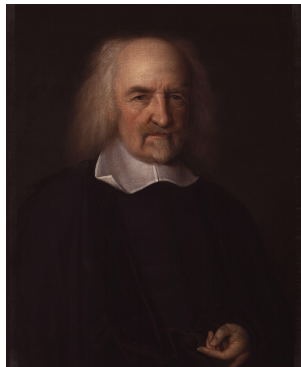
In strategic settings, destroying your own options
can make you better off.

- ▶ A **game**: players + individual strategies + payoffs.
- ▶ A **strategy**: what you would do under different circumstances.
- ▶ **Payoffs** are rank outcomes. The numbers are not dollars or utils.
- ▶ **Strategic behavior**: my best choice depends on what I expect **you** to do.

Philosopher	His state of nature	His game
Hobbes	the war of all against all	Prisoner's Dilemma
Locke	inconvenient: every man judge in his own case	Hawk-Dove
Rousseau	peaceful, but trust is fragile	Stag Hunt

- ▶ Not a history of how states arose, a model of what enforcement is for.
- ▶ “But the state of nature never existed”: it is not supposed to have.

- ▶ Born 5 April 1588 at Westport, Wiltshire.
- ▶ *Leviathan* published 1651, written in exile.
- ▶ Died 4 December 1679, aged 91.

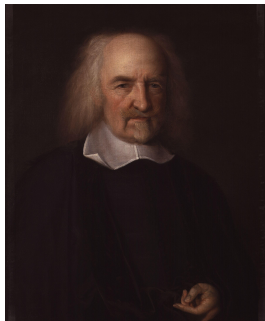


Thomas Hobbes (1588–1679)

- ▶ In nature, every man has a right to everything $\Rightarrow$ the paradox of universal rights: where all have a right to all, no one securely has anything.
- ▶ No common power $\Rightarrow$ nothing keeps anyone in awe $\Rightarrow$ war of every man against every man.
- ▶ War is a condition, not a battle: the standing disposition to fight is enough to poison everything else.

Hobbes, *Leviathan*, ch. XIII, 1651

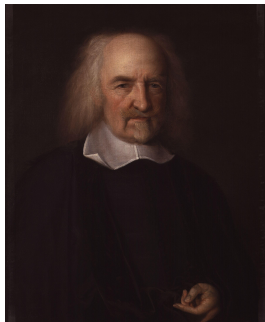
"Hereby it is manifest, that during the time men live without a common Power to keep them all in awe, they are in that condition which is called Warre; and such a warre, as is of every man, against every man."



Thomas Hobbes
(1588–1679)

Hobbes, *Leviathan*, ch. XIII, 1651

"In such condition, there is no place for Industry; because the fruit thereof is uncertain; and consequently no Culture of the Earth; no Navigation, nor use of the commodities that may be imported by Sea; no commodious Building; no Instruments of moving, and removing such things as require much force; no Knowledge of the face of the Earth; no account of Time; no Arts; no Letters; no Society; and which is worst of all, continuall feare, and danger of violent death; And the life of man, solitary, poore, nasty, brutish, and short."



Thomas Hobbes
(1588–1679)

Three motives that lead to the war of all against all:

- ▶ **Competition**: attack for gain and to make yourself master of other men's persons and goods.
- ▶ **Diffidence**: attack for safety, to strike before you are struck.
- ▶ **Glory**: attack for reputation.

Three motives that lead to the war of all against all:

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- ▶ **Glory**: attack for reputation.

- ▶ Two neighbours in the state of nature. Each chooses: **Attack** or **Refrain**.
- ▶ Each person ranks the options as follows:
 1. plunder an unarmed neighbour
 2. mutual peace
 3. mutual war
 4. be the one plundered

Neighbour 2 Neighbour 1	Refrain	Attack
Refrain	3, 3	0, 5
Attack	5, 0	1, 1

Neighbour 2 Neighbour 1		Refrain	Attack
		Refrain	Attack
Refrain	3, 3	0, 5	
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Diagram illustrating the Hobbesian game payoff matrix. The matrix shows the outcomes for two players, Neighbour 1 and Neighbour 2, based on their choices of Refrain or Attack. The payoffs are given as (Neighbour 1, Neighbour 2).

Payoffs:

- (Refrain, Refrain): 3, 3
- (Refrain, Attack): 0, 5
- (Attack, Refrain): 5, 0
- (Attack, Attack): 1, 1

Arrows indicate the following transitions:

- A blue arrow points from (Refrain, Refrain) to (Refrain, Attack).
- Red arrows point from (Refrain, Refrain) to (Attack, Refrain) and from (Refrain, Attack) to (Attack, Attack).

Neighbour 2		Refrain	Attack
Neighbour 1			
Refrain		3, 3 	0, 5 
Attack		5, 0 	1, 1 

Neighbour 2		Refrain	Attack
Neighbour 1			
Refrain		3, 3 → 0, 5	
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- (Refrain, Attack): 0, 5
- (Attack, Refrain): 5, 0
- (Attack, Attack): 1, 1

Arrows indicate transitions from the (Refrain, Refrain) and (Attack, Refrain) outcomes to the (Attack, Attack) outcome. The (Attack, Attack) outcome is highlighted with a red box.

- ▶ Attack beat Refrain against a refrainer ($5 > 3$) and against an attacker ($1 > 0$): it is best **whatever the other does**.
- ▶ Such a strategy is **dominant**; the strategy it always beats is **dominated**, and a rational player never uses one.
- ▶ Both neighbours reason the same way $\Rightarrow$ both attack. No wickedness was assumed anywhere: only the ranking of outcomes.

- ▶ A **Nash equilibrium**: a pair of strategies where no player can gain by changing **alone**.
 - ▶ (Attack, Attack) qualifies as the Nash equilibrium in the hobbesian game.
 - ▶ Peace (Refrain, Refrain) does **not**: each neighbour gains by deviating.

Even people who would all prefer peace
end up at war.

- ▶ Leviathan: a common power to keep them all in awe, armed with punishment.
- ▶ If the penalty exceeds the temptation, Refrain becomes the dominant strategy, and peace stops needing virtue.

Neighbour 2		
Neighbour 1	Refrain	Attack
	Refrain	Attack
Refrain	3, 3	0, 1
Attack	1, 0	-3, -3

Punishment of 4 on every attack, enforced with certainty.

Neighbour 2		Neighbour 1	
		Refrain	Attack
Neighbour 1	Refrain	3, 3	0, 1
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Neighbour 2		Refrain	Attack
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Refrain		3, 3	0, 1
Attack		1, 0	-3, -3

The table shows the following payoffs:

- If both Refrain: (3, 3)
- If Neighbour 1 Refrains and Neighbour 2 Attacks: (0, 1)
- If Neighbour 1 Attacks and Neighbour 2 Refrains: (1, 0)
- If both Attack: (-3, -3)

Arrows indicate best responses for each player given the other's strategy:

- A blue arrow points from (0, 1) to (3, 3), indicating that Neighbour 2's best response to Neighbour 1's Refrain is to Refrain.
- Two red arrows point from (1, 0) and (-3, -3) to (3, 3), indicating that Neighbour 1's best response to Neighbour 2's Refrain or Attack is to Refrain.

Punishment of 4 on every attack, enforced with certainty.

Neighbour 2		Refrain	Attack
Neighbour 1			
Refrain		3, 3	0, 1
Attack		1, 0	-3, -3

Punishment of 4 on every attack, enforced with certainty.

		Neighbour 2	
		Refrain	Attack
Neighbour 1	Refrain	3, 3	0, 1
	Attack	1, 0	-3, -3

The image shows a 2x2 payoff matrix for a game between Neighbour 1 and Neighbour 2. The strategies are Refrain and Attack. The payoffs are (Neighbour 1, Neighbour 2). The (Refrain, Refrain) cell has a payoff of (3, 3) and is highlighted with a red box. The (Refrain, Attack) cell has a payoff of (0, 1). The (Attack, Refrain) cell has a payoff of (1, 0). The (Attack, Attack) cell has a payoff of (-3, -3). Red arrows point from the (Attack, Refrain) and (Attack, Attack) cells to the (Refrain, Refrain) cell, indicating a punishment of 4 on every attack.

Punishment of 4 on every attack, enforced with certainty.





- ▶ **Arms races**: building is dominant, and both powers end poorer and no safer.
- ▶ **Cartels**: cheating on the quota is dominant $\Rightarrow$ cartels unravel from inside.
- ▶ **Doping**: if the others dope, you must. If they don't, doping wins anyway. Everyone dopes, rankings unchanged.

Cheung, "The Contractual Nature of the Firm," 1983

"My own favorite example is riverboat pulling in China before the communist regime, when a large group of workers marched along the shore towing a good-sized wooden boat. The unique interest of this example is that the collaborators actually agreed to the hiring of a monitor to whip them."



Trackers above Ichang, c. 1910

- ▶ The pullers' problem: every man prefers all-pull to all-shirk, and each is tempted to let the others pull.
- ▶ Their solution: **hire the whip themselves**. The monitor is chosen, paid, and dismissible by the monitored.
- ▶ Cheung's whip is a purchased solution: a Leviathan in miniature, **chosen by contract**.

- ▶ Discount factor $\delta \in (0, 1)$: patience + the probability the interaction continues.
- ▶ A payoff of 3 tomorrow is worth 3δ today; in two days, $3\delta^2 \Rightarrow$ the future is real, but it trades at a discount.

$t = 0$

	Refrain	Attack
Refrain	3, 3	0, 5
Attack	5, 0	1, 1

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	Refrain	Attack
Refrain	3, 3	0, 5
Attack	5, 0	1, 1

$t = 1$

	Refrain	Attack
Refrain	$3\delta, 3\delta$	0, 5δ
Attack	$5\delta, 0$	δ, δ

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- ▶ A payoff of 3 tomorrow is worth 3δ today; in two days, $3\delta^2 \Rightarrow$ the future is real, but it trades at a discount.

$t = 0$

	Refrain	Attack
Refrain	3, 3	0, 5
Attack	5, 0	1, 1

$t = 1$

	Refrain	Attack
Refrain	$3\delta, 3\delta$	0, 5δ
Attack	$5\delta, 0$	δ, δ

$t = 2$

	Refrain	Attack
Refrain	$3\delta^2, 3\delta^2$	0, $5\delta^2$
Attack	$5\delta^2, 0$	δ^2, δ^2

...

- ▶ Begin by refraining. Refrain as long as the other refrains.
- ▶ If you are **ever** attacked: attack forever. No appeals, no second chances.

Cooperate forever:

$$V_C = \sum_{t=0}^{\infty} 3\delta^t = 3 + 3\delta + 3\delta^2 + \dots + 3\delta^{\infty}$$

$$V_C = 3 + \delta V_C$$

$$V_C = \frac{3}{1 - \delta}$$

UF | When is cooperation an equilibrium?

Cooperate forever:

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$$V_C = \frac{3}{1 - \delta}$$

Attack today and face war ever after:

$$V_A = 5 + \sum_{t=1}^{\infty} \delta^t$$

$$V_A = 5 + \frac{\delta}{1 - \delta}$$

UF | When is cooperation an equilibrium?

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Attack today and face war ever after:

$$V_A = 5 + \sum_{t=1}^{\infty} 1\delta^t$$

$$V_A = 5 + \frac{\delta}{1 - \delta}$$

Cooperation holds if and only if

$$\frac{3}{1 - \delta} \geq 5 + \frac{\delta}{1 - \delta} \iff \delta \geq \frac{1}{2}$$

- ▶ Repeated interaction + enough patience = cooperation is an equilibrium without a Leviathan.
- ▶ This is not utopian: it is how merchants, neighbours and nations behave wherever they expect to meet again.
- ▶ Everything turns on $\delta \Rightarrow$ whatever lengthens the shadow of the future makes peace more likely.

UF | What lengthens the shadow?

- ▶ **Repeated interaction**: the more often you meet, the more future matters.
- ▶ **Patience**: the more you value tomorrow, the more you fear tomorrow's retaliation.
- ▶ **Reputation**: if others know your past, they can punish you for defecting today.





Bruce Almighty (2003).

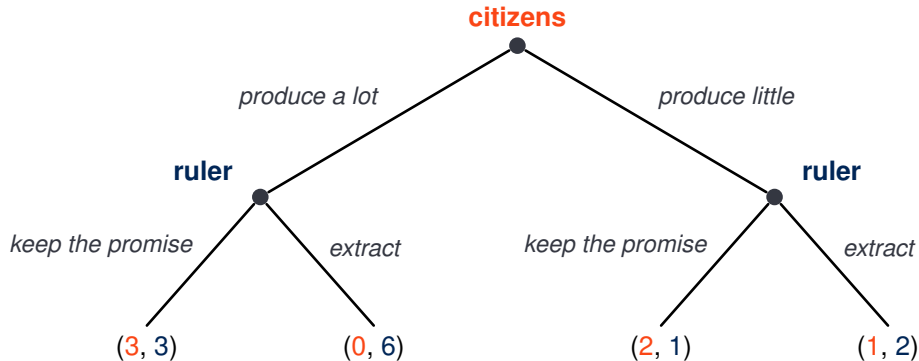
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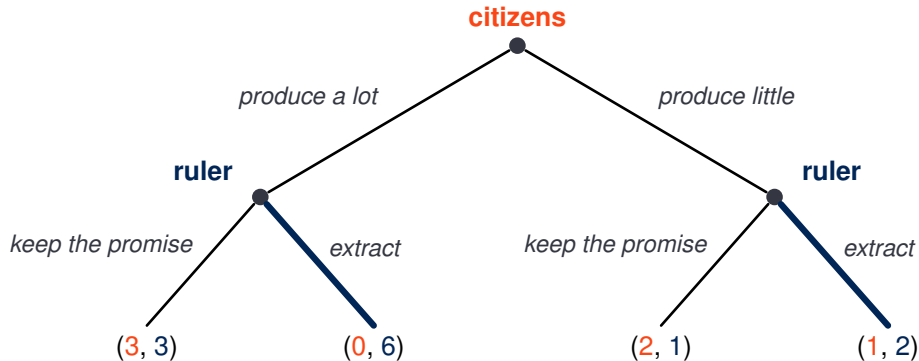
- ▶ Now let the end be **known**: the game is played exactly N times.
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- ▶ Round N : no future left $\Rightarrow$ defect.
- ▶ Round $N - 1$: round N is lost anyway $\Rightarrow$ defect. And so on, all the way back to today.
- ▶ With a known end, cooperation **unravels completely**. It does not matter whether N is 10 or 10,000.

- ▶ There is one thing an absolute ruler cannot dominate: **his own future will**.
- ▶ A sovereign above all law cannot bind himself $\Rightarrow$ there are promises he would **wish** to make and cannot make believed.
- ▶ So a prisoner's dilemma is reintroduced, this time between Leviathan and its subjects: the dilemma the state was built to solve reappears at the top.

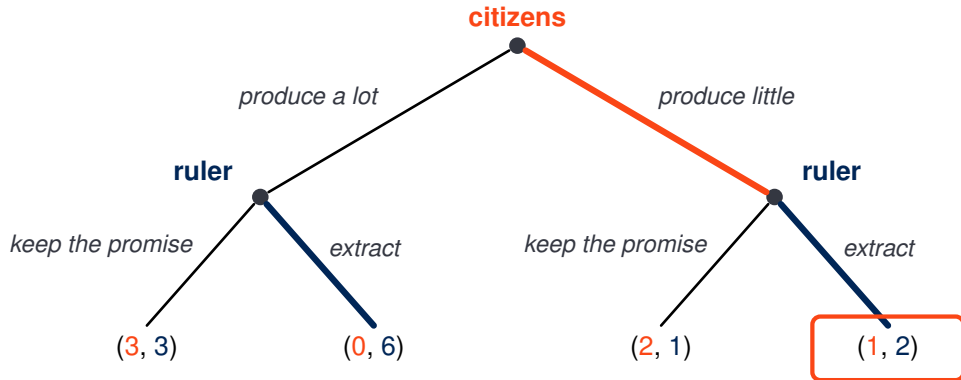


Output 9 if the citizens produce a lot, 3 if little. Producing a lot costs them 3 in effort. The promised tax takes $\frac{1}{3}$, extraction $\frac{2}{3}$: the same two rates in both worlds. Citizens' payoffs net of effort.



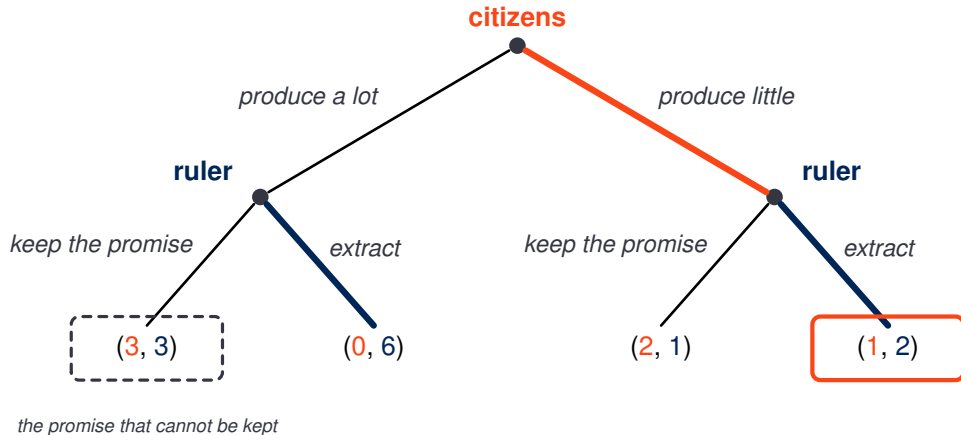
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UF | The ruler and the taxpayers



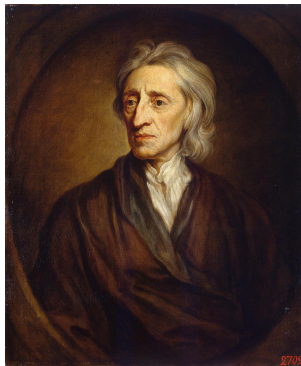
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Output 9 if the citizens produce a lot, 3 if little. Producing a lot costs them 3 in effort. The promised tax takes $\frac{1}{3}$, extraction $\frac{2}{3}$: the same two rates in both worlds. Citizens' payoffs net of effort.

- ▶ Born 29 August 1632 at Wrington, Somerset; his father a country lawyer who fought for Parliament in the civil war.
- ▶ Westminster School, then Christ Church, Oxford; trained in medicine.
- ▶ Fled to Holland in 1683 and returned in 1689, in the wake of the Glorious Revolution.
- ▶ *Two Treatises of Government* published anonymously, 1689. He owned the book only in his will. Died 28 October 1704.

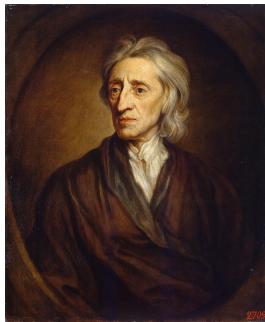


John Locke (1632–1704)

- ▶ Not Hobbes's war. A **law of nature** already binds there: no harm to another's life, health, liberty, or possessions.
- ▶ Property exists **before** government: mixing your labor with what no one owns makes it yours.
- ▶ The defect is smaller and stranger: the state of nature is merely **inconvenient**.

Locke, *Second Treatise of Government*, §6, 1689

“The state of nature has a law of nature to govern it, which obliges every one: and reason, which is that law, teaches all mankind, who will but consult it, that being all equal and independent, no one ought to harm another in his life, health, liberty, or possessions . . .”

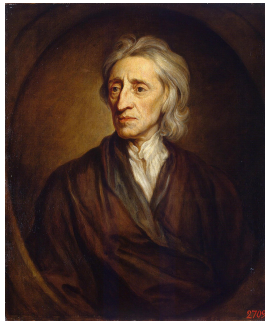


John Locke
(1632–1704)

- ▶ Every man is judge and executioner **in his own case**: each enforces the law of nature himself.
- ▶ Self-judged disputes tip into a state of war, and once begun, war between private enforcers **tends to continue**: no umpire exists to end it.
- ▶ So conflict is possible, not inevitable $\Rightarrow$ the right model must let **war and deference coexist**. That is not a prisoner's dilemma.

Locke, *Second Treatise of Government*, §19, 1689

“And here we have the plain difference between the state of nature and the state of war, which however some men have confounded, are as far distant, as a state of peace, good will, mutual assistance and preservation, and a state of enmity, malice, violence and mutual destruction, are one from another.”



John Locke
(1632–1704)

Claimant 2 Claimant 1		Dove	Hawk
		Dove	Hawk
Dove	3, 3	0, 4	
Hawk	4, 0	-1, -1	

Resource worth 6 when shared in peace. Arming costs 2, a fight costs 2 more, and each side wins it half the time.

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Claimant 2 \ Claimant 1		Dove	Hawk
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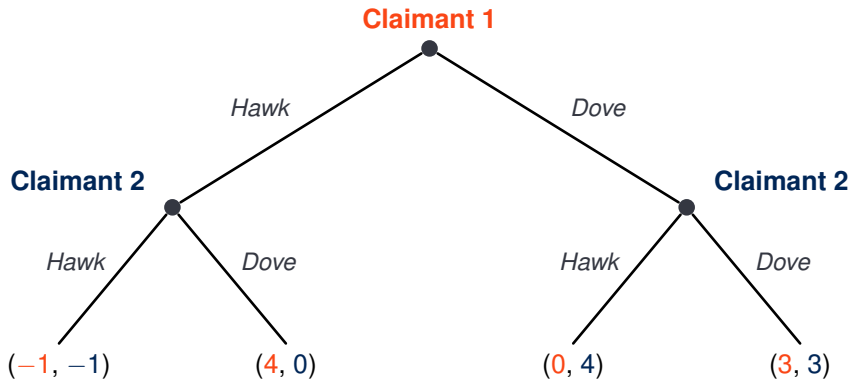
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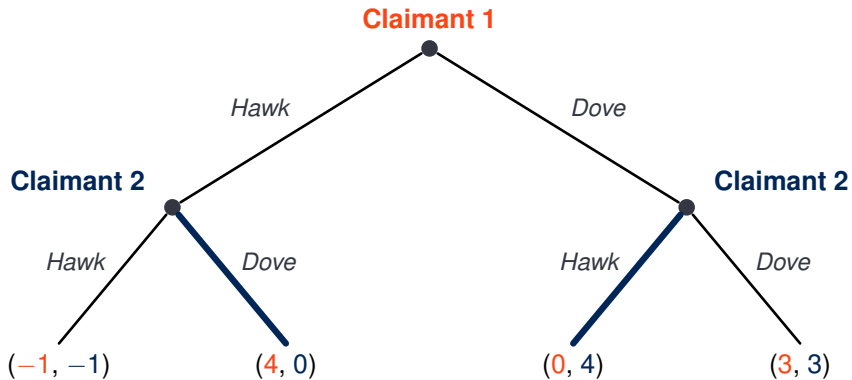
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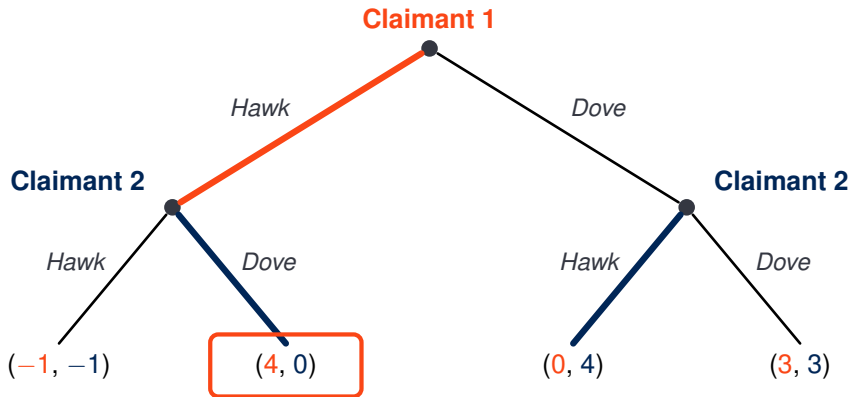
Resource worth 6 when shared in peace. Arming costs 2, a fight costs 2 more, and each side wins it half the time.

- ▶ No dominant strategy: the best reply **depends on what the other does**. $\Rightarrow$ Meet a Hawk, be a Dove. Meet a Dove, be a Hawk.
- ▶ Two pure equilibria: (Hawk, Dove) and (Dove, Hawk). Someone yields (the game does not say who).
- ▶ New fact of life: **the matrix alone no longer predicts**. Something outside it must select the equilibrium.

Source: Bruner, Justin P. (2020). "Locke, Nozick and the state of nature," *Philosophical Studies* 177, 705–726.





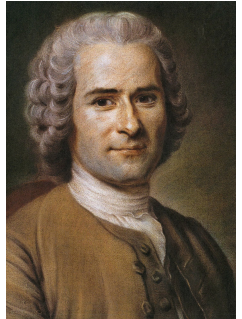


- ▶ Order the moves and the ambiguity vanishes: the first mover plays Hawk, the second concedes. **First-mover advantage**.
- ▶ The Lockean reading: **possession moves first**. The incumbent claims $\Rightarrow$ the latecomer defers.
- ▶ Property is the equilibrium in which fights do not happen: deference to possession, self-enforcing.
- ▶ Yet resources are lost relative to a world where all play “Dove”: 4 for the pair, not 6.

- ▶ Hawk-Dove is a workhorse of international relations: brinkmanship, crisis bargaining.
- ▶ In the mixed equilibrium, war occurs with **positive probability** although both players prefer any peaceful split.
- ▶ Conflict as equilibrium **randomness**, not equilibrium intention: nobody chose the war, and it is nobody's mistake.

Rousseau, *Discourse on the Origin of Inequality*, Part II, 1755

“If a deer was to be taken, every one saw that, in order to succeed, he must abide faithfully by his post: but if a hare happened to come within the reach of any one of them, it is not to be doubted that he pursued it without scruple, and, having seized his prey, cared very little, if by so doing he caused his companions to miss theirs.”



Jean-Jacques Rousseau
(1712–1778)

- ▶ Born 28 June 1712 in Geneva, son of a watchmaker. His mother died days after his birth.
- ▶ Largely self-taught. Earned his living as a music copyist. His opera *Le Devin du village* played before Louis XV in 1752.
- ▶ The *First Discourse* won the Academy of Dijon's prize in 1750. the *Discourse on Inequality* (1755), written for a later Dijon competition, did not win.
- ▶ *Émile* and *The Social Contract* published in 1762: *Émile* condemned and burned in Paris, both books burned in Geneva ⇒ arrest warrants drove him into years of flight.
- ▶ Died 2 July 1778 at Ermenonville ⇒ moved to the Panthéon by the National Convention, 1794.

- ▶ The inverse of Hobbes: natural man is solitary, self-sufficient, **peaceful**.
- ▶ His motives are **amour de soi**, self-preservation, and **pity**: not glory.
- ▶ Conflict is not natural: it is **acquired**. Society creates the things men fight over before it creates the peace to enjoy them.

Hunter 2 Hunter 1		Stag	Hare
		Stag	Hare
Stag	Hare	10, 10	0, 7
		7, 0	7, 7

Hunter 2 Hunter 1		Stag	Hare
Stag	10, 10	0, 7	
Hare	7, 0	7, 7	



Hunter 1 \ Hunter 2 Hunter 2	Stag	Hare
	Stag	Hare
Stag	10, 10	0, 7
Hare	7, 0	7, 7



Hunter 2 Hunter 1		Stag	Hare
		Stag	Hare
Stag	10, 10	0, 7	
Hare	7, 0	7, 7	

Hunter 1 \ Hunter 2		Stag	Hare
		Stag	Hare
Stag		10, 10	0, 7
Hare		7, 0	7, 7

Diagram illustrating the Stag Hunt game matrix. The matrix shows the payoffs for two hunters (Hunter 1 and Hunter 2) based on their choices of Stag or Hare. The payoffs are shown in red text. Blue arrows indicate the best response for each hunter given the other's choice. Red arrows indicate the best response for each hunter given the other's choice.

- If Hunter 1 chooses Stag and Hunter 2 chooses Stag, the payoff is (10, 10).
- If Hunter 1 chooses Stag and Hunter 2 chooses Hare, the payoff is (0, 7).
- If Hunter 1 chooses Hare and Hunter 2 chooses Stag, the payoff is (7, 0).
- If Hunter 1 chooses Hare and Hunter 2 chooses Hare, the payoff is (7, 7).

Blue arrows indicate the best response for each hunter given the other's choice:

- From (Stag, Stag) to (Stag, Hare): Hunter 1's best response to Stag is Stag.
- From (Stag, Hare) to (Hare, Hare): Hunter 1's best response to Hare is Hare.
- From (Hare, Stag) to (Stag, Stag): Hunter 2's best response to Stag is Stag.
- From (Hare, Hare) to (Stag, Hare): Hunter 2's best response to Hare is Hare.

Hunter 2 Hunter 1		Stag	Hare
		Stag	Hare
	Stag	10, 10	0, 7
	Hare	7, 0	7, 7

The diagram illustrates the Stag Hunt game matrix. The rows represent Hunter 1's strategies (Stag, Hare) and the columns represent Hunter 2's strategies (Stag, Hare). The payoffs are shown in the cells. Red boxes highlight the two Nash equilibria: (Stag, Stag) with payoffs (10, 10) and (Hare, Hare) with payoffs (7, 7). Blue arrows indicate best responses: from (Stag, Stag) to (Stag, Stag) and from (Hare, Hare) to (Hare, Hare). Red arrows indicate best responses: from (Stag, Hare) to (Hare, Hare) and from (Hare, Stag) to (Stag, Stag).

- ▶ Two equilibria, both self-enforcing: (Stag, Stag) and (Hare, Hare).
- ▶ (Stag, Stag) is **payoff-dominant**: better for everyone. (Hare, Hare) is **risk-dominant**: 7 guaranteed, trust required of no one.
- ▶ The deepest contrast with the prisoner's dilemma: there the bad outcome was **forced**. Here nothing forces either. The question has changed from “why defect?” to “**which equilibrium?**”

- ▶ Hunt the stag only if you believe your partner does: it pays when $10q \geq 7$, i.e. when you are at least 70% sure of him.
- ▶ Not “some” trust: a lot of trust. At fifty-fifty, hunt the hare.
- ▶ Cooperation here is fragile, not impossible (contra Hobbes) and not self-enforcing (contra Locke’s optimism).

- ▶ The prisoner's dilemma asks: why would anyone ever cooperate?
- ▶ The stag hunt asks the older and better question: **how do we get from the hare equilibrium to the stag equilibrium?**
- ▶ This is the problem the social contract tries to solve.

- ▶ The stag equilibrium hangs on **beliefs about beliefs**: I hunt stag only if I believe you believe I will.
- ▶ Whatever anchors those expectations selects the equilibrium: norms, ritual, language, reputation, focal points.
- ▶ That is **social capital** in the strict sense: the stock of things that select the good equilibrium.



Hume, *A Treatise of Human Nature*, III.ii.2, 1740

“Two men, who pull the oars of a boat, do it by an agreement or convention, tho’ they have never given promises to each other.”



David Hume
(1711–1776)

- ▶ No promise was ever exchanged, yet the stroke settles: a **convention**, an equilibrium each expects the other to keep.
- ▶ Most of social life runs on such equilibria: money, language, which side of the road.
- ▶ Conventions solve stag hunts **without** anyone in charge. But they can be bought, too.

- ▶ Each gives himself to the **general will**, and receives in exchange the assurance that everyone else hunts the stag.
- ▶ Freedom is not lost but transformed: **obeying a law you prescribe to yourself**.
- ▶ Note what the contract does **not** do: it does not change the payoffs. It **selects the equilibrium**.

	The problem	The solution
Hobbes	war is the only equilibrium	an absolute sovereign changes the payoffs
Locke	every man judge in his own case	limited government enforces pre-existing rights
Rousseau	two equilibria, fragile trust	the general will selects the good equilibrium

- ▶ Two players, two stages: a **one-shot** prisoner's dilemma first, then a stag hunt. Then the game ends.
- ▶ Finite, known end $\Rightarrow$ cooperation should unravel under a prisoner's dilemma.
- ▶ The question: can the hunt **discipline** the dilemma?

1 \ 2	Coop.	Defect
Coop.	3, 3	0, 5
Defect	5, 0	1, 1

stage 1: the dilemma

then →

1 \ 2	Stag	Hare
Stag	10, 10	0, 7
Hare	7, 0	7, 7

stage 2: the hunt

“Defect today, and tomorrow we hunt hare.”

- ▶ The strategy: cooperate, then hunt stag; if you defect, I hunt hare. Absolutely. No matter what.

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- ▶ The strategy: cooperate, then hunt stag; if you defect, I hunt hare. Absolutely. No matter what.
- ▶ Why believe it? (Hare, Hare) is itself an equilibrium of the stage-2 subgame: carrying out the punishment is a best reply, not spite.
- ▶ So nothing unravels: unlike the repeated dilemma, the last stage has two equilibria to choose between. Punishment works by equilibrium selection.

- ▶ Cooperate, then stag: $3 + 10 = 13$.

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- ▶ Defect, then hare: $5 + 7 = 12$.

- ▶ Cooperate, then stag: $3 + 10 = 13$.
- ▶ Defect, then hare: $5 + 7 = 12$.
- ▶ $13 > 12 \Rightarrow$ cooperating in the dilemma is part of a **subgame-perfect equilibrium**:
temptation ($5 - 3 = 2$) is beaten by the equilibrium-selection punishment ($10 - 7 = 3$).

- ▶ A society facing both a cooperation problem and a coordination problem can use the second to **discipline the first**.
- ▶ Hobbes's problem, partly solved with Rousseau's game: **social trust as the enforcement device**.
- ▶ No Leviathan anywhere in the construction.

Is anarchy so bad?

next class